

GENERAL SPECIFICATION G5.21

Earthing System

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G5.21 EARTHING SYSTEM

G5.21.1 General

(a) Introduction

(i) General

- 1) This Section is intended to generally cover the design, supply, deliver, installation, testing and commissioning for the earthing system. The schematic earthing drawings (if provided) are only indicative and contractor shall design the complete earthing system in compliance with SS 551 duly endorsed by LEW, with appropriate Engineer Grade.
- 2) The earthing system shall comprise the earth electrodes, main earthing terminal bar(s), earthing busbar(s), earthing conductors, main earthing tinned copper tape network, electrodes, earthing chambers equipotential bonding conductors and circuit protective conductors.
- 3) The design of earthing system shall satisfy the following requirements:
 - Adequate thermal capacity to withstand the system prospective fault current for the duration for which the protective systems are designed to clear the fault; and
 - The earth connection for all sections of each installation shall be continuous throughout.

(b) References

- (i) The following is a list of standards which may be referenced in this Section. The Contractor also shall refer to General Specification G5.2 - Standards and Regulations:

1) British Standards Institute (BSI):

Standard	Subject
BS 951	Electrical Earthing. Clamps for Earthing and Bonding Specification.
BS 5733	Specification for General Requirements for Electrical Accessories.
BS 6121	Mechanical Cable Glands. Code of Practice for Selection, Installation and Inspection of Cable Glands Used in Electrical Installations
BS 7430	Code of Practice for Earthing.

(ii) International Electrotechnical Commission (IEC)

Standard	Subject
IEC 60050-195	International Electrotechnical Vocabulary – Part 195: Earthing and Protection Against Electric Shock.
IEC 60364	Low Voltage Electrical Installations.
IEC/TR 61000-5-2	Electromagnetic Compatibility (EMC) – Part 5: Installation and Mitigation Guidelines – Section 2: Earthing and Cabling.
IEC 61643-1	Surge Protective Devices Connected to Low Voltage Power Distribution Systems – Part 1: Performance Requirements and Testing Methods.

(iii) Singapore Productivity and Standards Board (PSB)

Standard	Subject
CP 5	Code of Practice for Electrical Installations.
SS 551	Code of Practice for Earthing.
SS 555	Code of Practice for Lightning Protection.
SS 322	Specification for Earthing and Bonding Clamps.

(c) Submittals

(i) Shop Drawings, and Technical Data: As a minimum, submit the following items:

1) Product Data

- Exothermic weld connectors;
- Exothermic welding procedure;
- Mechanical connectors;
- Compression connectors;
- Earth electrodes;
- Earth Bar and busbar;
- Earth pit; and
- Signal reference grid.

G5.21.2 Products

- (a) Earth Electrode
 - (i) Material: Individual 16 mm diameter copper-clad steel rod with a radial thickness of 0.25 mm of 99.9 percent pure electrolytic copper molecularly bonded onto a steel core. Coupling shall be of long length high copper alloy; and
 - (ii) Length: 3.6 metre, minimum.
- (b) Main Earthing Terminal Bar and Earthing Busbar
 - (i) 50 mm x 6.5 mm (main earthing bar), 25 mm x 3 mm (earthing tap) shall be hard drawn, high conductivity tinned copper bar;
 - (ii) Length as required;
 - (iii) The main earth bar (Complete with test link) shall be installed at a convenient location on the switchroom wall or other location as required, 600 mm above the finished floor. Earth stud shall have a minimum size of M8;
 - (iv) The earthing terminal bar shall be provided with adequate number of terminal connection points for connection to the earth bar or connection point at each transformer, switchgear, motor control centre and distribution panel in the area in addition to other connections shown on the schematic Drawings;
 - (v) The earthing bar and frame of each apparatus shall be earthed directly to this terminal bar by means of a copper earthing conductor of adequate size to suit the current rating of the apparatus concerned;
 - (vi) Provide 20 percent spare terminal points complete with cable lugs and bolts; and
 - (vii) Provide an engraved plate with the earth electrode resistance value achieved and the date of testing. Securely install it near the main earthing terminal bar. The tested earth electrode resistance values shall be recorded and documented in the as-built drawings.
- (c) Earthing, Bonding, and Protective Conductors
 - (i) Bonding and Protective Conductors: As specified in the general specifications; and
 - (ii) Earthing Conductors
 - 1) 25 mm x 3 mm high conductivity, soft annealed, tinned copper tape;
 - 2) Copper tape shall be supplied in long, unbroken lengths. No joints shall be allowed in any run;
 - 3) Use cables if no joint in copper tape is possible;
 - 4) Cables when used shall comply with the general specifications; and
 - 5) The earthing system shall be electrically continuous and shall meet the requirements of SS CP 5, SS 551, IEC and BS standards.

- (d) Connectors
 - (i) Exothermic Weld Type
 - 1) Outdoor Weld: Suitable for exposure to elements or direct burial; and
 - 2) Indoor Weld: Utilise low-smoke, low-emission process.
 - (ii) Compression Type
 - 1) Meeting applicable requirements of BS 4579 or equivalent IEC;
 - 2) Compress-deforming type; wrought copper extrusion material;
 - 3) Single indentation for conductors 16 mm² and smaller;
 - 4) Double indentation with extended barrel for conductors 25 mm² and larger; and
 - 5) Barrels pre-filled with oxide-inhibiting and anti-seizing compound and sealed.
 - (iii) Mechanical Connections
 - 1) Split-bolt, saddle, or cone screw type; copper alloy material; and
 - 2) Bolts, screws, nuts, washers and rivets for copper conductors shall be phosphor-bronze, naval brass, or copper silicon.
 - (iv) Protective Coatings
 - 1) Solvent cutback thixotropic corrosion preventative forming a film of resilient matt petroleum wax;
 - 2) Fast drying durable rubberised spray coating; and
 - 3) Heat shrink clear sheathing.
- (e) Earthing Electrode Inspection Pit
 - (i) Inspection pit shall be concrete or extremely high corrosion resistance and of long life span, UV stable, shatter-proof, chemical resistance polymer earth pit of approved design. Capable of withstanding the vehicle load of 10,000 kg for pit located at driveway and capable of withstanding 5,000 kg for pit located at non-driveway. The cover shall be marked EARTH ELECTRODE INSPECTION PIT.
 - (ii) Earth Electrode Inspection Pit shall extend to a depth of 500 mm below finished ground level and keep free of soil.
 - (iii) Provide an engraved, corrosion resistant labels for each electrode inspection pit and shall be marked on the as-built drawing.
 - (iv) Each earth electrode shall be clearly marked 'SAFETY ELECTRICAL EARTH CONNECTION – DO NOT REMOVE'.
- (f) Marker Post: Marker posts shall be provided to mark the position of the electrodes adjacent to the earth chambers. The marker post shall be similar to those provided for cable routes.

- (g) Surge Diverter between LV and Instrument earthing
 - (i) Designed to provide equipotential bonding under transient conditions;
 - (ii) Automatic reset after operation;
 - (iii) Surge Rating: 100 kA 8/20 microsecond surge rating;
 - (iv) Breakdown Voltage: 230 volts;
 - (v) Rated Life: 10,000 operations, minimum; and
- (h) Raised Floor Earthing
 - (i) Provide and install signal reference grid and connectors for raised floor earthing as specified on schematic drawings and / or standard details.

G5.21.3 Execution

- (a) General
 - (i) Earthing shall be in compliance with SS 551 and CP 5;
 - (ii) The main earth bar in each building shall be mounted 600 mm above finished floor;
 - (iii) Bond together system neutrals, service equipment enclosures, exposed noncurrent-carrying metal parts of electrical equipment, metal raceways, earth conductor in raceways and cables, socket earth connections, and metal piping systems;
 - (iv) Shielded Power Cables: Earth shields at each splice or termination in accordance with recommendations of splice or termination manufacturer; and
 - (v) Shielded Instrumentation Cables
 - 1) Earth shield to earth bus at power supply for analogue signal;
 - 2) Expose shield minimum 25 mm at termination to field instrument and apply heat shrink tube; and
 - 3) Do not earth instrumentation cable shield at more than one point.
- (b) Wire Connections
 - (i) Circuit Protective Conductors: Install in raceways containing power conductors and control circuits above 50 volts;
 - (ii) Non-Metallic Raceways and Flexible Tubing: Install equipment circuit protective conductor connected at both ends to noncurrent-carrying earthing bus;
 - (iii) Connect circuit protective conductors to raceway earthing bushings;
 - (iv) Extend and connect circuit protective conductors to earth bus in all equipment containing an earth bus;
 - (v) Connect (bond) enclosure of equipment containing earth bus to that bus;
 - (vi) Bolt connections to equipment earth bus;
 - (vii) Connect (bond) earth conductors to metallic enclosures at each end, and to intermediate metallic enclosures; and

- (viii) Junction Boxes: Supply and install materials and connect to equipment earthing system with connectors approved by the S.O..
- (c) Equipment Earthing
 - (i) Extend earth system in each electrical room to equipment served via protective conductors installed in raceway, cable tray, or cable ladder. Connect to equipment frame.
 - (ii) All non-current carrying metal framework associated with electrical equipment, each item of the electrical plant, casings and cable sheaths etc., shall be earthed by means of tinned copper or copper conductors to the earthing network comprising of earth electrodes to be run around the perimeter of each location in a manner approved by the S.O.. The serial connection of the earthing of one major equipment to another equipment casing without another parallel earthing path will not be permitted.
 - (iii) Circuit and main earthing and equipotential bonding conductors shall be high conductivity tinned copper tape or 1000 V grade PVC insulated multi-stranded cable. PVC cable insulation shall be striped green/yellow. Cable length shall be continuous and intermediate jointing is not permitted.
 - (iv) The main bonding cable shall not be less than 16 sq.mm and supplementary bonding of non-electrical plant shall not be less than 10 sq.mm. All connections shall be made using compression lugs, taped on completion to complete seal the lug and any bare copper from the atmosphere.
 - (v) The earth network conductor system comprising tinned copper tapes and /or cables shall be securely supported and clamped in a neatly spaced manner on building structures, cable trench, and laid on cable trays, neatly run vertically, horizontally, parallel to adjacent walls, ceilings, beam, etc. The earthing installation shall be protected from mechanical damage or corrosion.
 - (vi) Where cable tray or cable ladder is used as the protective conductor, supply and install an appropriately sized conductor from the end of the tray or ladder to the equipment.
 - (vii) Non-Metallic Raceways and Flexible Tubing: Supply and install a circuit protective conductor connected at both ends to noncurrent-carrying earth bus.
 - (viii) Motors Less Than 10 kW: Supply and install compression, spade-type terminal connected to equipment cable termination box mounting screw.
 - (ix) Motors 10 kW and Above: Tap equipment frame or housing; supply and install compression, one-hole, lug type terminal connected with minimum M8 brass bolt and washer.
 - (x) Equipment Circuits 20 Amperes or Above: Tap equipment frame housing; supply and install solderless terminal with minimum M8 bolt.
- (d) Earth Electrodes
 - (i) Install full length with conductor connection at upper end;
 - (ii) Install with connection point below finished grade, unless otherwise shown; and
 - (iii) Space multiple electrodes by 3.6 metre, minimum.

- (e) Earthing Electrode Inspection Chamber
 - (i) Install outside buildings, asphalt, and paved areas;
 - (ii) Install cover flush with surface; and
 - (iii) Place 250 mm crushed rock in bottom of each chamber.
- (f) Connections
 - (i) General
 - 1) The matting surface of all tapes at joints, terminals, etc., shall be tinned before clamping;
 - 2) All contact surfaces shall be thoroughly cleaned and coated with an anti-corrosive electrical jointing compound suitable for the conductor material;
 - 3) All connections to electrical apparatus and earth electrodes shall be bolted;
 - 4) Remove paint, dirt, or other surface coverings at connection points to allow good metal-to-metal contact; and
 - 5) Underground joints and connections shall be protected by a coating which will form a seal and exclude moisture. The coating shall cover all exposed conductors and the material it is connected to;
 - 6) Notify S.O. prior to backfilling earth connections.
 - (ii) Connections shall be made as follows:
 - 1) Phosphor-bronze set screws at main earthing bar(s);
 - 2) Bronze or copper clamps with phosphor-bronze bolts to earth electrodes. Edges of clamps shall be rounded;
 - 3) Phosphor-bronze bolts and nuts for earth connection to pipes. Bolt directly to the flange of the pipe;
 - 4) Connections of flat strip; two M8 bolts or four 5 mm diameter rivets;
 - 5) Connection to sheet metal; two M8. Where sheet metal is less than 2 mm thick it shall be backed for an area of at least 1,000 mm²;
 - 6) Riveting or exothermic welding to earth plates; and
 - 7) Exothermic welds to steel reinforcing steel or steel piles.
 - (iii) Exothermic Weld Type
 - 1) Wire brush or file contact point to bare metal surface;
 - 2) Use welding cartridges and moulds in accordance with manufacturer's recommendations;
 - 3) Avoid using badly worn moulds;
 - 4) Mould to be completely filled with metal when making welds; and
 - 5) After completed welds have cooled, brush slag from weld area and thoroughly clean joint.

- (iv) Compression Type
 - 1) Install in accordance with connector manufacturer's recommendations;
 - 2) Install connectors of proper size for earthing conductors and earth electrodes specified; and
 - 3) Install using connector manufacturer's compression tool having proper sized dies.
- (v) Mechanical Type
 - 1) Apply homogeneous blend of colloidal copper and rust and corrosion inhibitor before making connection;
 - 2) Install in accordance with connector manufacturer's recommendations; and
 - 3) Do not conceal mechanical connections.
- (g) Metal Structure Earthing
 - (i) Earth metal sheathing and exposed metal vertical structural elements to earth electrode system;
 - (ii) Bond electrical equipment supported by metal platforms to the platforms; and
 - (iii) Provide electrical contact between metal frames and railings supporting pushbutton stations, receptacles, and instrument cabinets, and raceways carrying circuits to these devices.
- (h) Draw Pit Earthing
 - (i) Install one earth electrode inside each Draw Pit;
 - (ii) Electrode Floor Protrusion: 100 to 150 mm above floor;
 - (iii) Make connections of protective conductors fully visible and accessible; and
 - (iv) Connect all noncurrent-carrying metal parts, and any metallic raceway to earth electrode with 16 mm² conductor.
- (i) Transformer Earthing
 - (i) Bond neutrals of transformers within buildings to building earth system, and to any additional indicated earthing electrodes;
 - (ii) Bond neutrals of large, oil-filled transformers to earthing system;
 - (iii) Bond neutrals of substation transformers to substation earthing system.
- (j) Raised Floor Earthing
 - (i) Install and bond control equipment and as recommended by equipment supplier; and
 - (ii) Provide additional earthing as directed by S.O..
- (k) Surge Protection Equipment Grounding
 - (i) Connect surge arrestor earth terminals to equipment earth bus.

- (l) Testing
 - (i) The work shall also include testing of all electrodes to the satisfaction of the S.O..
 - (ii) After the earthing installation has been completed, the Contractor shall demonstrate to the S.O. that the resistance of the electrodes to earth and continuity of the earth network are within the limits specified. Earth tests shall include measurement of resistance value of each individual earth point. The result of all earth resistance measurements shall be certified by the Contractor's LEW.
 - (iii) Earth resistance testing shall be carried out once in the driest period of the year or any time deemed suitable by the S.O. during the Defects Liability Period. Should the earth resistance value less than the specified value, the Contractor shall install additional earthing points in order to achieve the specified value. The Contractor's rate shall be deemed to include such additional earthing points without claims for additional cost in this respect.
 - (iv) The Contractor shall produce earthing schematic drawings including the location of earth pits and connections to switchboards, MCCs, DBs, transformers, etc.

END OF SECTION